reelforge: feature overview

Script-to-animated-reel platform. Context document for ChatGPT, 4 Nov 2026. Private repo: github.com/Aayushvz/reelforge

1. What it is

A creator writes a short script, adds voice lines, and gets back a finished, cinematic, lip-synced animated short for Instagram Reels and Shorts. The defaults are 9:16 at 1080×1920; 16:9, 3:4 and 1:1 are planned. The goal is relatable, emotional, high-retention videos with recurring characters. Token and API costs are kept low: Claude makes at most one small call per video, and everything else runs locally.

2. Core pipeline (built and working)

- **Script / performance file**: each line has a speaker, emotion (11 types), intensity, mood, pause and punchline flag.
- **Voice**: ElevenLabs. Lines are generated in the web app and imported as mp3s, or generated through the API. Voices carry emotion tags such as [sighs] and [excited]. Output is cached so lines are never paid for twice.
- **Timing**: local faster-whisper alignment gives word- and letter-level timings. A timeline and dialogue mix are built from them.
- **Facial animation**: 52 ARKit face channels in layers. An emotion layer with natural overshoot, lip sync (mouth shapes timed to the audio), blinks (faster when nervous), and head nods on stressed words.
- **Automatic director**: wide, medium and close-up shots; close-ups on punchlines; cuts every 3 s or less; push-ins and handheld drift.
- **Mood lighting**: each story beat has a mood (calm, success, panic, awkward, reveal, sad). Lights and glowing screens crossfade between moods.
- **3D render**: headless Blender 4.5 (Eevee with ray tracing), with depth-of-field bokeh, haze and AgX colour.
- **Finish**: highlight glow (halation), grade, vignette, film grain, and burned-in captions. Captions highlight each word as it's spoken, with a colour per character and a pop and wobble on the punchline. Audio is normalised to -14 LUFS.
- **CLI**: reelforge build-assets and reelforge run <project> (quick stills preview or full render). 49 automated tests.
- **Optional**: an NVIDIA Audio2Face adapter (runs locally on CPU), plugged in per line.

3. Multi-style platform

Each video picks an animation **style engine**. The script, voice, timing, captions, sound and editor are shared; only the picture-making stage is swapped.

Style	Look	Engine / cost	Status
blender3d	Stylized 3D characters in moody, glowing sets (@canwebuildthis / NMXWZ vibe)	Blender, free, local	Pipeline built; cast and sets next
cinematic-ai	Pixar-like cinematic shots: lantern streets, aerials, VFX (ink.industries look)	AI keyframes -> image-to-video -> lip sync. Paid providers are opt-in (owner's key); free fallback is Wan 2.2	Planned
Styles 3–6	From future references	TBD	Future

4. Character cast (planned)

Seven **original** characters in a stylized-sculpt look: caricature proportions, sculpted clay-like hair, soft skin, an expressive face rig. Real-person references are used for style only.

Character	Vibe	Signature look
Kai (M)	chill creative, deadpan	tight black curls, chain, earring
Rey (M)	loud, big laugh	curly hair, full beard, glasses
Dev (M)	confident coder, smirk	fade with swoop, stubble, watch
Mo (M)	anxious overthinker	messy hair, round glasses
Aria (F)	witty lead, sarcastic	long dark hair, hoops
Zoe (F)	bubbly, dramatic reactions	high ponytail, freckles, blush
Noor (F)	calm, stylish, dry humour	sleek bob or braids, gold jewellery

- **Gen-Z wardrobe system**: slots for top, outerwear, bottoms, shoes and accessories; 25+ items; recolourable; at least 3 outfits per character.
- Each character ships with a turnaround, an expression sheet (laugh, smug, surprised, sad, angry, blink, talk) and a talking test clip.

5. Environment library (planned)

Ten sets, each with 3 lighting presets (30 looks in total), built as depth-mapped cinematic backdrops plus real 3D props and lights.

Set	Lighting presets
Night-city desk	blue night · neon pink · rainy
Bedroom / gaming room	warm lamp · RGB LEDs · morning sun
Rooftop skyline	golden hour · blue hour · night lights
Café window	rainy · sunny afternoon · fairy lights
Lantern old-town street	sunset · lantern night · foggy dawn
Subway platform	fluorescent · late-night empty · rush-hour warm
Campus / classroom	daylight · overcast · after-hours
Startup office	daylight · late-night crunch · spotlight
Night convenience store	fluorescent · neon sign · rain outside
Beach pier	sunset · pastel morning · carnival lights

6. Editor app (planned)

A local web app in dark mode, clean and minimal. Higgsfield and InVideo are references for feel and flow only; the design language is our own. No AI-slop patterns: no purple gradients, glassmorphism, emoji icons or sparkles.

- **Screens**: Projects -> Style picker -> Script -> Cast (wardrobe drawer) -> Set (lighting chips) -> Voice -> Editor -> Approve stills -> Render/Export.
- **Main editor**: preview player in the centre, a shot list on the left with per-shot regenerate, an inspector on the right (camera, mood, captions, expressions), and a multi-track timeline below (dialogue, shots, moods, captions, music/SFX). Keyboard-first.
- **Design system**: near-black neutrals, one warm amber accent, Inter or Geist type, an 8 px grid, hairline borders, Lucide icons, subtle motion.
- **Tech**: Vite + React + TypeScript, a Three.js 3D preview, and a local backend wrapping the reelforge CLI.

7. Built for views

- A hook in the first 1.5 s: optionally open cold on the strongest line.
- Loopable endings, recurring characters (series identity), and emotional beats shown through voice, face, body, lighting and camera.

- Sound design: foley, ambience, a music bed that ducks under dialogue, a stinger on the punchline (planned).
- Script feedback from Claude (weak hook, slow setup), at about 300 extra tokens.

8. Constraints and decisions

- No paid tools for the Blender style. Paid AI video is opt-in for the cinematic-ai style only.
- Claude never creates accounts or handles passwords or API keys. The user signs in themselves.
- Local-first: an RTX 4060 laptop (8 GB), Blender 4.5, ffmpeg, Python 3.12+.
- Workflow: brainstorm -> spec -> plan -> build with reviewed tasks; honest quality reviews with renders.

9. Roadmap

- 0. Engine refactor (pluggable styles)
- 1. Character cast and wardrobe
- 2. 10 environments × 3 lighting presets
- 3. Editor UI
- 4. cinematic-ai engine
- 5. More styles from new references